

Module 09: Database Indexes and Access Paths

Database Indexes Architecture & B-Tree Access Paths

Clustered vs Non-Clustered Indexes, Dense vs Sparse Indexing & Query Tuning

1. Clustered vs Non-Clustered Index

Clustered Index (Primary Data Ordering)

- Data rows are **PHYSICALLY** sorted on disk in index key order!
- Leaf nodes contain the **ACTUAL DATA ROWS** (no pointers needed).
- **Exact ONE Clustered Index allowed per table (data can only be sorted one way!).**
- Auto-created when you declare a **PRIMARY KEY**.

```
CREATE CLUSTERED INDEX idx_roll ON Student(RollNo);
```

Non-Clustered / Secondary Index

- Physical data is **NOT** re-sorted. Index is a separate auxiliary structure.
- Leaf nodes contain Index Key + Row Locator (RID / Pointer to data page).
- **Multiple Non-Clustered Indexes allowed per table (e.g., up to 999).**

```
CREATE INDEX idx_name_city ON Student(Name, City); -- Composite
```

Trade-off: Fast search $O(\log N)$ vs slower INSERT/UPDATE (index maintenance).

2. Dense vs Sparse Indexing

Dense Index

- Index record exists for **EVERY SINGLE** search key value in table.
- Faster search (direct pointer to target record without linear block scan).
- Consumes larger memory and storage space.

Can be built on both sorted and unsorted data files.

Sparse Index (Block Anchors)

- Index record exists only for **SOME** search key values (e.g. 1 per disk block).
- Much smaller index size (easily fits in main memory RAM cache).
- **Strict Requirement: The underlying data file MUST be physically sorted!**

Search: Follow index to block anchor → perform linear scan inside block.

Ideal for primary clustered indexes on massive enterprise datasets.

Indexing Trade-off: Indexes drastically accelerate SELECT query retrieval from $O(N)$ to $O(\log N)$, but add overhead to INSERT/UPDATE/DELETE statements due to index tree maintenance.

1. Clustered vs Non-Clustered Index

Feature	Clustered Index	Non-Clustered Index
Physical Sort	Physically sorts table data on disk.	Independent B-Tree pointing to data rows.
Count Limit	Exactly 1 per table.	Multiple allowed per table.
Leaf Nodes	Actual data rows.	Search keys + Row Pointers.

2. Sparse Indexing Rule

A Sparse Index stores block anchors (1 entry per block). It is compact enough to fit in RAM, but strictly requires the underlying data file to be sorted.